QUESTION 1.

B) The dataset from the mass package is called crabs. It contains 200 observations on 9 variables, providing morphological measurements for 50 crabs from each of 4 groups:

Two species (blue B and orange O) and both sexes (male M and female F). The measurements include frontal lobe size FL, rear width RW , carapace length CL, carapace width CW, and the body depth BD.

D) Yes, the dataset is balanced. It contains exactly 50 specimens for each of the four possible combinations: blue male, blue female, orange male, orange female.

QUESTION 2

B) Outliers are rare for the bd variable

Central tendency

The orange has a higher median BD than the blue species.

Males typically show a higher median BD compared to females.

The orange male exhibit the highest central tendency for bd.

The males have an overall greater variation, because they have higher “whiskers”.

QUESTION 5.

B) Strength: variables show very strong positive linear correlations.

Nature: the relationships are linear. This means a straight line effectively describes how one measurement changes relative to another.

QUESTION: 6

INTRODUCTION

The study examines how FL, RW, CL, CW and BD differ across 2 species (blue and orange) and both sexes. The goal is to determine if these physical dimensions can reliably distinguish between these biological groups

The initial bd through side-by-side boxplot reveals distinct differences in size and variability.

The distribution of carapace length visualized through histograms show a relatively symmetric shape for both species.

The carapace width shows the most significant variation between species.

With sex variation, the RW measurement shows the least variation between sexes.

The scatterplot matrix reveals very strong positive linear relationship between all 5 variables.

The relationships are strictly linear with very little scatter away from the main trend lines.

Correlation coefficients are near 0.9

Conclusion

The statistical analysis of the crabs dataset suggests that multivariate models could highly accurately classify crabs into their respective species and sexes.